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EEE 313 — Past Questions & Materials
by Renance

Course: EEE 313
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200, 300

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Paper 1 of 3 — EEE 313 - 2024/2025 - First Semester - Electromagnetic Field and Waves I

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
 First Semester Examination 2024/2025 Session
 Course Title: Electromagnetic Field and Waves I Course Code: EEE 313
 Time Allotted: 3 Hours Course Unit: 3

Instruction: Answer five (5) questions. All assumptions should be clearly stated and please do not write at the front of the question paper. You can do your rough works at the back of your booklet.

QUESTION ONE

Given that $E = 3ax + 4ay + 5az$, $Q = 2ax - 6ay + 7az$, $U = \rho x \sin \phi + z \cos \phi + \rho^2$ and $V = 4\rho \sin \theta \cos \phi$, determine the following:

- (i) The angle (θ) between E and Q .
- (ii) $\text{Div } U = \nabla \cdot U$
- (iii) $\text{Grad } U = \nabla U$
- (iv) $\nabla^2 V$.

(14 marks)

QUESTION TWO

Given the moving loop in Figure 1, If $B = 0.6a_z \text{ Wb/m}^2$, $R = 10 \Omega$, $L = 10 \text{ cm}$ and the rod is moving with a constant velocity of $8a_x \text{ m/s}$, find:

- (i) The induced emf (V_{emf}) in the rod.
- (ii) The current (i) through the resistor.
- (iii) The motional force (F_m) on the rod.
- (iv) The power (P) dissipated by the resistor.

(14 marks)

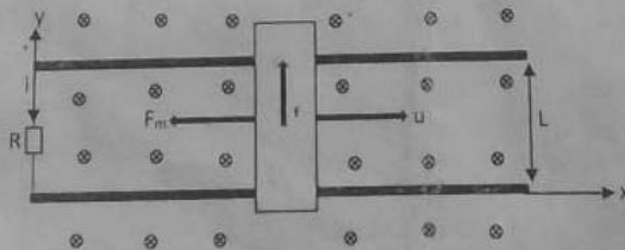
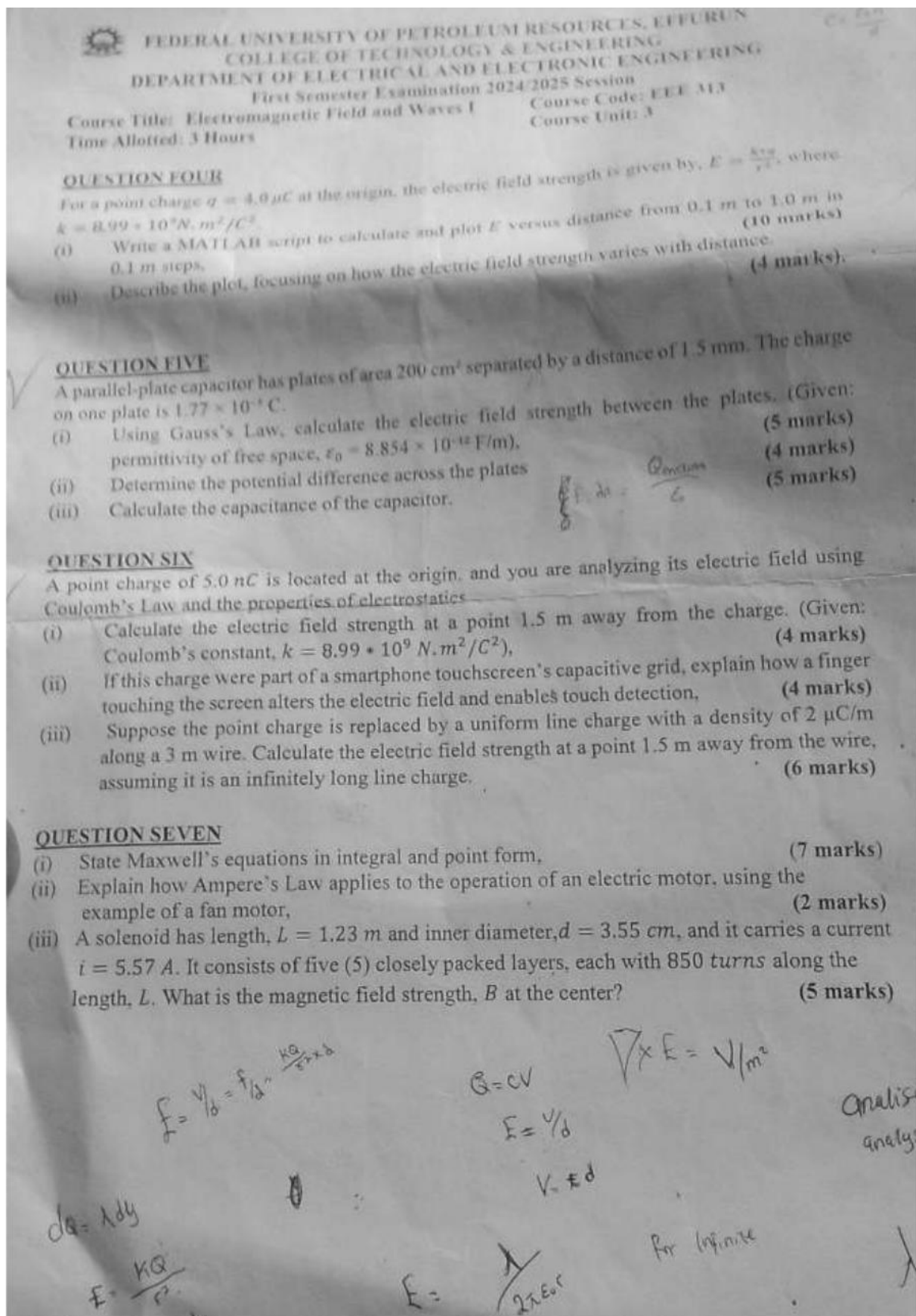


Figure 1: Induced emf due to moving loop in a static B field

QUESTION THREE

Within a certain region the permeability of the medium (ϵ) = 10^{-12} F/m and the permittivity of the medium (μ) = 10^{-6} H/m . If the flux density (B_z) = $2 \times 10^{-4} \sin 10^4 t \cos 10^{-3} y \text{ T}$: Use $\nabla \times H = \epsilon \frac{dE}{dt}$ to determine the electric field (E) of the medium.

(14 marks)



Paper 2 of 3 — EEE 313 Electromagnetic Field & Waves I 2025

Level 300 • First Semester • PastQuestion

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Emf - Qv.B


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COLLEGE OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
First Semester Examination 2024/2025 Session
Course Title: Electromagnetic Field and Waves I **Course Code: EEE 313**
Time Allotted: 3 Hours **Course Unit: 3**

Instruction: Answer five (5) questions. All assumptions should be clearly stated and please do not write at the front of the question paper. You can do your rough works at the back of your booklet.

A.B

QUESTION ONE

Given that $E = 3ax + 4ay + 5az$, $Q = 2ax - 6ay + 7az$, $U = \rho x \sin \phi + z \cos \phi + \rho^2$ and $V = 4r \sin \theta \cos \phi$, determine the following:

- The angle (Θ) between E and Q,
- $\text{Div} U = \nabla \cdot U$
- $\text{Grad} U = \nabla U$
- $\nabla^2 V$.

(14 marks)

QUESTION TWO

Given the moving loop in Figure 1, If $B = 0.6a_z \text{ Wb/m}^2$, $R = 10 \Omega$, $L = 10 \text{ cm}$ and the rod is moving with a constant velocity of $8a_x \text{ m/s}$, find:

- The induced emf (V_{emf}) in the rod,
- The current (i) through the resistor,
- The motional force (F_m) on the rod,
- The power (P) dissipated by the resistor.

(14 marks)

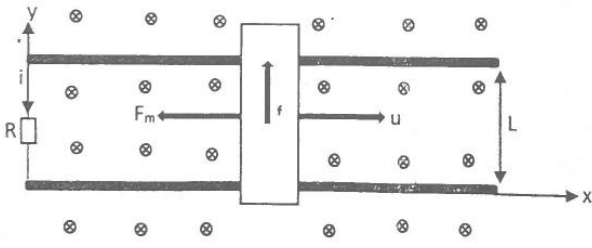


Figure 1: Induced emf due to moving loop in a static B field

QUESTION THREE

Within a certain region the permeability of the medium (ϵ) = 10^{-12} F/m and the permittivity of the medium (μ) = 10^{-6} H/m . If the flux density (B_x) = $2 \times 10^{-4} \sin 10^4 t \cos 10^{-3} y \text{ T}$: Use $\nabla \times H = \epsilon \frac{dE}{dt}$ to determine the electric field (E) of the medium.

(14 marks)



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Course Title: Electromagnetic Field and Waves I
Time Allotted: 3 Hours

Course Code: EEE 313
Course Unit: 3

$$C = \frac{\epsilon_0 A}{d}$$

QUESTION FOUR

For a point charge $q = 4.0 \mu\text{C}$ at the origin, the electric field strength is given by, $E = \frac{k \cdot q}{r^2}$, where $k = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$.

- Write a MATLAB script to calculate and plot E versus distance from 0.1 m to 1.0 m in 0.1 m steps. (10 marks)
- Describe the plot, focusing on how the electric field strength varies with distance. (4 marks).

QUESTION FIVE

A parallel-plate capacitor has plates of area 200 cm^2 separated by a distance of 1.5 mm . The charge on one plate is $1.77 \times 10^{-5} \text{ C}$.

- Using Gauss's Law, calculate the electric field strength between the plates. (Given: permittivity of free space, $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$), (5 marks)
- Determine the potential difference across the plates (4 marks)
- Calculate the capacitance of the capacitor. (5 marks)

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

QUESTION SIX

A point charge of 5.0 nC is located at the origin, and you are analyzing its electric field using Coulomb's Law and the properties of electrostatics.

- Calculate the electric field strength at a point 1.5 m away from the charge. (Given: Coulomb's constant, $k = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$), (4 marks)
- If this charge were part of a smartphone touchscreen's capacitive grid, explain how a finger touching the screen alters the electric field and enables touch detection, (4 marks)
- Suppose the point charge is replaced by a uniform line charge with a density of $2 \mu\text{C/m}$ along a 3 m wire. Calculate the electric field strength at a point 1.5 m away from the wire, assuming it is an infinitely long line charge. (6 marks)

QUESTION SEVEN

- State Maxwell's equations in integral and point form, (7 marks)
- Explain how Ampere's Law applies to the operation of an electric motor, using the example of a fan motor, (2 marks)
- A solenoid has length, $L = 1.23 \text{ m}$ and inner diameter, $d = 3.55 \text{ cm}$, and it carries a current $i = 5.57 \text{ A}$. It consists of five (5) closely packed layers, each with 850 turns along the length, L . What is the magnetic field strength, B at the center? (5 marks)

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{KQ}{r^2} \times \frac{KQ}{r^2}$$

$$dQ = \lambda dy$$

$$E = \frac{KQ}{r^2}$$

$$E = \frac{K\lambda dy}{r^2}$$

$$Q = CV$$

$$E = \frac{V}{d}$$

$$V = Ed$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

For infinite

For finite

Analyses
analyses

1/2R

Paper 3 of 3 — EEE 313 Past Question(2019-2021)

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUM
COLLEGE OF ENGINEERING AND TECHNOLOGY

EXAMINATION: First Semester 2020/2021 COURSE CODE: EEE 313

SESSION: 2020/2021 COURSE TITLE: Electromagnetic Field and Theory I

DEGREE: B. Eng. TIME ALLOWED: 3 Hours

DATE: November 15, 2021

INSTRUCTION: Answer Question One and any other four.

Take $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, electronic charge, $e = -1.6 \times 10^{-19} \text{ C}$.

Question One

- Find Cartesian coordinates of point P whose cylindrical coordinates are $(12, \pi/6, 5)$.
- Write expressions for each following in Cartesian coordinate system.
 - Differential displacement, $d\mathbf{l}$,
 - differential normal surface areas, ds and
 - differential volume, dv .
- Give mathematical expressions for the vectors R_1 , R_2 , and D shown in figure 1c
- Find the unit vector of D , i.e a_D from 1c

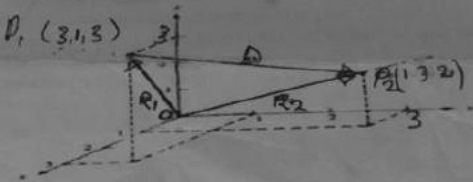


Figure 1

- Write the integral form of Maxwell's equation and identify each equation with its experimental law.
- Corresponding to a scalar field $V = 2xy - 5z$, is a vector field given by $E = \left(\frac{\partial V}{\partial x}\right) \mathbf{a}_x + \left(\frac{\partial V}{\partial y}\right) \mathbf{a}_y + \left(\frac{\partial V}{\partial z}\right) \mathbf{a}_z$. Find E at the point $(1, 2, 3)$.

Question Two

- State Gauss's law (b). State the condition that favors the use Gauss's law in determining the charge enclosed in a volume.
- Find the charge in the volume defined by $0 \leq x \leq 2$, $0 \leq y \leq 2$, and $0 \leq z \leq 2$ if $\rho_v = 30x^2y \text{ (}\mu\text{C/m}^3\text{)}$

1

Socratic

d. A cube of 2-m side is centered at the origin with the edges parallel to the axes. Within the cube, $D = \left(\frac{10x^2}{3}\right)az$ (C/m²). By application of Gauss's law, determine the total charge contained in the cube? Note that since the cube is centered at the origin, the edges run from -1 to +1.

Question Three

a. State Coulomb's law (b). A 100 μ C point charge (Q_1), is located in a rectangular coordinate system at (1, 1, 1) m, and another point charge (Q_2), of 50 μ C, is at (-1, 0, -2) m. Find the vector force on the first charge. (c). Find (i) the electric field intensity E, at (0, 0, 10) m due to two charges, $Q_1 = 0.35 \mu$ C at (0, 8, 0) m and $Q_2 = -0.55 \mu$ C at (6, 0, 0) m. (ii) Also find the electric flux density D at the same point.

Question Four

(a). State the Ampere's circuit law. (b) Find a current distribution that produces a magnetic field of the form $H = 100 \sin x a_y$. (c). In a cylindrical coordinate system, the magnetic flux density $B = \left(\frac{8.0}{\rho}\right) a_\phi$. Determine the magnetic flux ϕ , crossing the plane surface defined by $0.1 \leq \rho \leq 2$ m and $0 \leq z \leq 4$ m.

Question Five

(a). State Faraday's law (b) Discuss the three conditions under which electromotive force (emf) could be produced with reference to Faraday's law. (c) A circular loop of 10-cm radius is located in the xy plane in a B-field given by $B = (0.5 \cos 377t)(3a_y + 4a_z)$. Determine the voltage induced in the loop. The loop is defined by $\phi = 2\pi$.

Question Six

(a) What do you understand by the dielectric constant of a material? (b) Briefly explain what you understand by conduction and convection currents (c) 'A perfect conductor has infinite conductivity ($\sigma = \infty$) and cannot contain an electrostatic field within it'. Discuss (d). The cable shown in Fig. 2 is 8 km long. Calculate its capacitance.

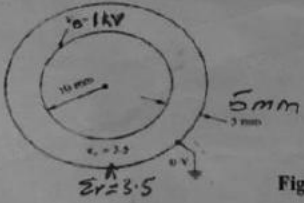


Fig 2

Question Seven

(a) An electric field is given by $E = xy^2 a_x + xz^2 a_y + xyz a_z$. Calculate the x component of its curl at (2, 1, 3). (b) Evaluate the divergence of the E-field of Problem 7(a) at (1, 2, 3). (c) A point charge of 1 μ C is located at the origin of a cylindrical coordinate system. Determine the potential difference between two points 1 m and 2 m radially away from the origin. (d) (a) Given a vector function, $E = 4y a_x + 5x a_y$. Find the scalar line integral $\int E \cdot dl$ from P_1 (2, 1, -1) to P_2 (8, 2, -1).

2

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY

EXAMINATION: First Semester

COURSE CODE: EEE 313

SESSION: 2020/2021

COURSE TITLE: Electromagnetic Field and Theory I

DEGREE: B. Eng.

TIME ALLOWED: 3 Hours

DATE: November 15, 2021

INSTRUCTION: Answer Question One and any other four.

Take $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, electronic charge, $e = -1.6 \times 10^{-19} \text{ C}$.

Question One

- Find Cartesian coordinates of point P whose cylindrical coordinates are $(12, \pi/6, 5)$.
- Write expressions for each following in Cartesian coordinate system.
(i) Differential displacement, $d\mathbf{l}$, (ii) differential normal surface areas, $d\mathbf{s}$ and (iii) differential volume, $d\mathbf{v}$.
- Give mathematical expressions for the vectors $\mathbf{R}_1$, $\mathbf{R}_2$, and $\mathbf{D}$ shown in figure 1c
- Find the unit vector of $\mathbf{D}$, i.e. $\mathbf{a}_D$ from 1c

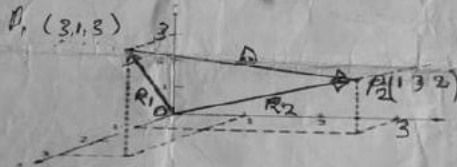


Figure 1

- Write the integral form of Maxwell's equation and identify each equation with its experimental law.

- Corresponding to a scalar field $V = 2xy - 5z$, is a vector field given by $\mathbf{E} = \left(\frac{\partial V}{\partial x}\right) \mathbf{a}_x + \left(\frac{\partial V}{\partial y}\right) \mathbf{a}_y + \left(\frac{\partial V}{\partial z}\right) \mathbf{a}_z$. Find $\mathbf{E}$ at the point $(1, 2, 3)$.

Question Two

- State Gauss's law (b). State the condition that favors the use Gauss's law in determining the charge enclosed in a volume.
- Find the charge in the volume defined by $0 \leq x \leq 2$, $0 \leq y \leq 2$, and $0 \leq z \leq 2$ if $\rho_v = 30x^2y \text{ } (\mu\text{C/m}^3)$

d. A cube of 2-m side is centered at the origin with the edges parallel to the axes. Within the cube, $D = \left(\frac{10x^2}{3}\right)az$ (C/m²). By application of Gauss's law, determine the total charge contained in the cube? Note that since the cube is centered at the origin, the edges run from -1 to +1.

Question Three

a. State Coulomb's law (b). A 100 μC point charge (Q_1), is located in a rectangular coordinate system at (1, 1, 1) m, and another point charge (Q_2), of 50 μC , is at (-1, 0, -2) m. Find the vector force on the first charge. (c). Find (i) the electric field intensity E , at (0, 0, 10) m due to two charges, $Q_1 = 0.35 \mu\text{C}$ at (0, 8, 0) m and $Q_2 = -0.55 \mu\text{C}$ at (6, 0, 0) m. (ii) Also find the electric flux density D at the same point.

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Question Six

(a) What do you understand by the dielectric constant of a material?

(b) Briefly explain what you understand by conduction and convection currents

(c) 'A perfect conductor has infinite conductivity ($\sigma = \infty$) and cannot contain an electrostatic field within it'. Discuss.

(d). The cable shown in Fig. 2 is 8 km long. Calculate its capacitance.

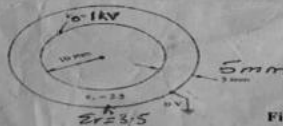


Fig 2

Question Seven

(a) An electric field is given by $E = xy^2 a_x - xz^2 a_y + xyza_z$. Calculate the x component of its curl at (2, 1, 3).

(b) Evaluate the divergence of the E -field of Problem 7(a) at (1, 2, 3).

(c) A point charge of 1 μC is located at the origin of a cylindrical coordinate system. Determine the potential difference between two points 1 m and 2 m radially away from the origin.

(d) (a) Given a vector function, $E = 4ya_x + 5xa_y$. Find the scalar line integral $\int E \cdot dl$ from P_1 (2, 1, -1) to P_2 (8, 2, -1). *229*

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
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EXAMINATION: First Semester

COURSE CODE: EEE 313

SESSION: 2019/2020

COURSE TITLE: Electromagnetic Field and Theory I

DEGREE: B. Eng

TIME ALLOWED: 3 Hours

DATE: March 22, 2021

INSTRUCTION: Answer Question One and any other four.

Take $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, electronic charge, $e = -1.6 \times 10^{-19} \text{ C}$.

Question One

- Write expressions for each following in cylindrical coordinate system.
(i) Differential displacement, $d\mathbf{l}$, (ii) differential normal surface areas, $d\mathbf{s}$ and (iii) differential volume, $d\mathbf{v}$.
- Given $\mathbf{A} = 2\mathbf{a}_x + 4\mathbf{a}_y$, and $\mathbf{B} = 6\mathbf{a}_x - 4\mathbf{a}_y$, find the smaller angle between them using (i) the cross product, (ii) the dot product.
- Find the distance between two points $A[5, 3\pi/2, 0]$ and $B[5, \pi/2, 10]$. Note that the two points are given in cylindrical coordinate system.
- Find the unit vector directed from $A(2, -5, -2) \text{ m}$ towards $B(14, -5, 3) \text{ m}$.
- Find the divergence of the vector field $\mathbf{P} = x^2yz\mathbf{a}_x + xz\mathbf{a}_z$.

Question Two

- State Coulomb's law.
- Two point charges, $Q_1 = 50 \mu\text{C}$ and $Q_2 = 10 \mu\text{C}$ are located at $(-1, 1, -3) \text{ m}$ and $(3, 1, 0) \text{ m}$ respectively as shown in figure 2a. Find the force on Q_1 .

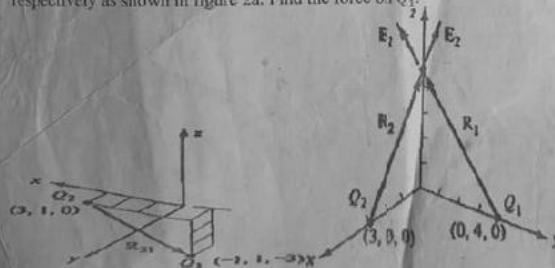


Fig. 2a

Fig. 2b

